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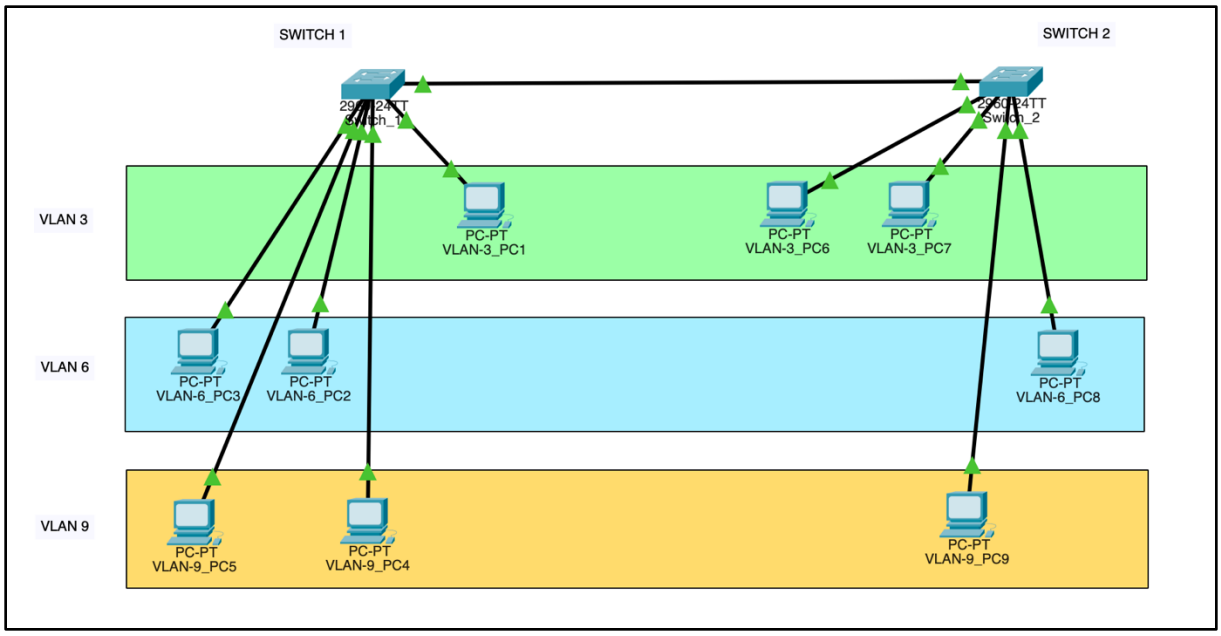
CEIT – 03 – 602P **February 14, 2026**

VLAN CONFIGURATION

SWITCH #	VLANs	Name	Default Gateway	PCs	IP Addresses	Ports
1	VLAN 3	Executives	192.168.3.1	PC1	192.168.3.10	Fa0/1
	VLAN 6	HR	192.168.6.1	PC2	192.168.6.10	Fa0/2
				PC3	192.168.6.11	Fa0/3
	VLAN 9	Finance	192.168.9.1	PC4	192.168.9.10	Fa0/4
				PC5	192.168.9.11	Fa0/5
2	VLAN 3	Executives	192.168.3.1	PC6	192.168.3.20	Fa0/6
				PC7	192.168.3.21	Fa0/7
	VLAN 6	HR	192.168.6.1	PC8	192.168.6.20	Fa0/8
	VLAN 9	Finance	192.168.9.1	PC9	192.168.9.20	Fa0/9

Network Topology



Network Connectivity

- **Same VLAN | SWITCH**

```
C:\>ping 192.168.9.10

Pinging 192.168.9.10 with 32 bytes of data:

Reply from 192.168.9.10: bytes=32 time=1ms TTL=128
Reply from 192.168.9.10: bytes=32 time=12ms TTL=128
Reply from 192.168.9.10: bytes=32 time=9ms TTL=128
Reply from 192.168.9.10: bytes=32 time=13ms TTL=128

Ping statistics for 192.168.9.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 13ms, Average = 8ms

C:\>
```

Ping PC5 from PC4 (SW 1 - VLAN 9)

```
C:\>ping 192.168.3.21

Pinging 192.168.3.21 with 32 bytes of data:

Reply from 192.168.3.21: bytes=32 time<1ms TTL=128
Reply from 192.168.3.21: bytes=32 time<1ms TTL=128
Reply from 192.168.3.21: bytes=32 time<1ms TTL=128
Reply from 192.168.3.21: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.3.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Ping PC7 from PC6 (SW 2 - VLAN 3)

- **Different SWITCH, same VLAN**

```
C:\>ping 192.168.6.10

Pinging 192.168.6.10 with 32 bytes of data:

Reply from 192.168.6.10: bytes=32 time=27ms TTL=128
Reply from 192.168.6.10: bytes=32 time<1ms TTL=128
Reply from 192.168.6.10: bytes=32 time<1ms TTL=128
Reply from 192.168.6.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.6.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 27ms, Average = 6ms

C:\>
```

Ping PC2 from PC8 (SW 1 from SW 2 - VLAN 6)

```
C:\>ping 192.168.9.20

Pinging 192.168.9.20 with 32 bytes of data:

Reply from 192.168.9.20: bytes=32 time=1ms TTL=128
Reply from 192.168.9.20: bytes=32 time<1ms TTL=128
Reply from 192.168.9.20: bytes=32 time<1ms TTL=128
Reply from 192.168.9.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.9.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Ping PC9 from PC4 (SW 2 from SW 1 - VLAN 9)

- **Same SWITCH, different VLAN**

```
C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Ping PC1 from PC5 (SW 1 - VLAN 3 and VLAN 9)

```
C:\>ping 192.168.3.21

Pinging 192.168.3.21 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.3.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Ping PC7 from PC8 (SW 2 - VLAN 3 and VLAN 6)

- **Different SWITCH | VLAN**

```
C:\>ping 192.168.3.21

Pinging 192.168.3.21 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.3.21:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Ping PC7 from PC2 (SW 2 from SW 1 – VLAN 3 from VLAN 6)

```
C:\>ping 192.168.6.10

Pinging 192.168.6.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.6.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Ping PC2 from PC9 (SW 1 from SW 2 – VLAN 6 from VLAN 9)